Lightweight transformer architecture for sexism classification

Jonathan Santos da Silva - a67606

Thesis presented to the School of Technology and Management in the scope of the Master in Informatics.

Supervisors:

Prof. Rui Pedro Lopes

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Dedication

I dedicate this work to God for His blessings and guidance. I would also like to express my gratitude to my family, my father Manoel Firmino Rodrigues da Silva, my mother Maria Oselha dos Santos Silva, and my brother Juliano Santos da Silva, for their support, love, and guidance throughout my whole life. And finally, I dedicate this work to my girlfriend and my friends who supported me in this whole process.

Acknowledgment

This work was supported by . . .

Abstract

Automatic sexism detection in multimodal online content is limited by methods that rely on a single modality or collapse annotator disagreement into a single majority label, losing information about genuine ambiguity in subjective judgments. This dissertation presents a multimodal architecture for sexism detection that trains directly on annotator disagreement instead of discarding it. The system combines text and image encoders with annotator demographic information, and, for video content, adds audio and physiological signals (eye-tracking and heart rate) through a shared fusion layer. Separate output heads jointly predict binary detection, source intention, and fine-grained category labels as probability distributions. An annotator reliability layer estimates how consistent each annotator is during training and generalizes this to new annotators using demographic information, avoiding the need for fixed annotator identities. This two-stage design turns local, modality-specific signals into a single, well-calibrated prediction. Experimental validation on memes and videos shows that combining text, image, demographic, and biometric signals under this soft-label approach produces more robust and better-calibrated predictions than methods using a single modality or majority-vote labels, while remaining lightweight enough to train without large-scale computational resources.

Keywords: Sexism detection, Multimodal learning, Learning with disagreement, Soft labels, Annotator reliability, Memes, TikTok videos, Biometric signals, Annotator demographics.

Resumo

A detecção automática de sexismo em conteúdo multimodal é limitada por métodos que usam apenas uma modalidade ou que reduzem o desacordo entre anotadores a um único rótulo majoritário, perdendo informação sobre a ambiguidade real de julgamentos subjetivos. Esta dissertação apresenta uma arquitetura multimodal para detecção de sexismo que treina diretamente sobre o desacordo entre anotadores, em vez de descartá-lo. O sistema combina codificadores de texto e imagem com informações demográficas dos anotadores e, para vídeos, adiciona sinais de áudio e fisiológicos (rastreamento ocular e frequência cardíaca) por meio de uma camada de fusão compartilhada. Cabeças de saída separadas preveem, em conjunto, a detecção binária, a intenção da fonte e as categorias finas como distribuições de probabilidade. Uma camada de confiabilidade dos anotadores estima o quão consistente cada anotador é durante o treino e generaliza essa estimativa para novos anotadores usando dados demográficos, evitando a dependência de identidades fixas de anotadores. Esse desenho em duas etapas transforma sinais locais, específicos de cada modalidade, em uma predição única e bem calibrada. A validação experimental em memes e vídeos mostra que combinar sinais de texto, imagem, demografia e biometria sob essa abordagem de rótulos soft produz predições mais robustas e melhor calibradas do que métodos com uma única modalidade ou rótulos de voto majoritário, permanecendo leve o suficiente para treinar sem grandes recursos computacionais.

Palavras-chave: Detecção de sexismo, Aprendizado multimodal, Aprendizado com divergência, Soft labels, Confiabilidade de anotadores, Memes, Vídeos do TikTok, Sinais biométricos, Dados demográficos dos anotadores.

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Acronyms

AI Artificial Intelligence.

CLEF Conference and Labs of the Evaluation Forum.

CLIP Contrastive Language-Image Pre-training.

EEG Electroencephalography.

EM Expectation-Maximization.

ESTiG Escola Superior de Tecnologia e Gestão.

EXIST sEXism Identification in Social neTworks.

GPU Graphics Processing Unit.

ICM Information Contrast Measure.

IPB Instituto Politécnico de Bragança.

LeWiDi Learning with Disagreement.

LoRA Low-Rank Adaptation.

MACE Multi-Annotator Competence Estimation.

ML Machine Learning.

MoE Mixture of Experts.

NLP Natural Language Processing.

PEFT Parameter-Efficient Fine-Tuning.

SWA Stochastic Weight Averaging.

UTFPR Universidade Tecnológica Federal do Paraná.

XLM Cross-lingual Language Model.

Chapter 1

Introduction

This chapter introduces the research context, motivations and objectives of this dissertation on sexism detection under the LeWiDi paradigm. With the recent growth of social networks, hate speech has become a concerning problem across online platforms. Sexism is one of its most frequent forms, appearing not only in written text (e.g. tweets, post captions) but also in images, memes, and most recently, short videos, where meaning often depends on how different types of content relate between each other. Detecting this kind of content automatically is hard, since sexist content is not always explicit, and people with different backgrounds or experience can disagree on what should be considered sexist. This dissertation addresses this problem by proposing a multimodal system that learns from these disagreements instead of discarding them.

Section 1.1 establishes the social and technological context that motivates this research, examining how current approaches to sexism detection fall short in dealing with multimodal content and subjective annotation. Section 1.2 presents the main problem addressed by this work, framing the specific limitations that the proposed system aims to overcome. Section 1.3 presents the main objective and specific contributions of this work, detailing the multimodal architecture developed and its application to sexism detection in memes and videos. Finally, Section 1.4 outlines the organization of this dissertation, providing a roadmap for the chapters that follow.

1.1 Contextualization

The expansion of social media apps has changed how people communicate, share opinions, and interact with one another. This has been followed by increasing interest in applying artificial intelligence and natural language processing techniques to understand a monitor online content, specifically in domains where harmful behavior can spread quickly, such as the automated identification of hate speech and discrimination content.

In this scenario, the ability to correctly detect sexist content is essential for supporting content moderation and maintain safer online environments. However, contemporary social media content presents increasingly complex challenges: sexism is frequently expressed not through just text, but the interaction between text and images, as in memes, or across multiple media types such as short-form videos. Furthermore, what counts as sexist is not always agreed upon, since it depends on cultural, personal, and contextual factors that vary between individuals.

Traditional approaches to sexism and hate speech detection, typically based on unimodal text analysis and majority vote labeling, have proven to not well address this complexity. As highlighted by[1], early hate speech detection relied on surface-level textual features that fail to capture the more subtle and context-dependent ways discriminatory context is expressed. Furthermore,[2] demonstrate that meme classification requires genuine cross-modal reasoning, since image and text can carry contradictory or complementary meaning that neither modality conveys alone Additionally, collapsing annotator judgements into a single majority-vote discards the information carried by disagreement itself, which[3] and [4] show to be a meaningful signal rather than noise, particularly on subjective tasks such as sexism detection.

Multimodal learning combined with the Learning With Disagreement (LeWiDi) paradigm offers a promising solution to these challenges by enabling a system to jointly reason over different types of content while preserving the distribution of annotator opinions instead of collapsing to a single ground truth. Under this paradigm, a model trained on soft label distributions can express calibrated uncertainty on contested instance, rather than committing to an overconfident and potentially incorrect decision.

This combined strategy is particularly valuable for sexism detection in memes and short-form videos, where visual, textual, and, in some cases behavioral or physiological signals must be considered together, and where the demographic background of annotators can meaningfully influence how content is perceived and labeled. If left unaddressed, this subjectivity, together with the difficulty of fusing heterogeneous modalities, can result in systems that are either overconfident on ambiguous content or unable to generalize across different types of media.

By implementing a multimodal, demographic-aware system trained under the LeWiDi paradigm, it is possible to achieve more robust and better-calibrated sexism detection while remaining lightweight enough to train without large-scale computational resources. As explored throughout this dissertation, combining textual, visual, and, for video content, biometric evidence within a single soft-label objective allows local, modality-specific signals to be aggregated into a more reliable characterization of sexist content than approaches relying on isolated modalities or collapsed labels.

1.2 Problem Statement

Despite recent progress in multimodal learning and subjective label modeling, three problems remain insufficiently addressed when it comes to detecting sexism in memes and short-form videos.

First, most existing multimodal hate speech detection systems are designed for a single type of content, typically static images or short text, and do not generalize easily to short-form video, where meaning is distributed across visual, textual, and auditory signals that unfold over time [5]. Recent efforts have further extended this challenge by introducing physiological data collected while participants view sexist content, opening a largely unexplored connection between perceptual signals and computational subjectivity models [6]. Extending multimodal fusion beyond memes to video and physiological signals, without relying on large-scale pretrained vision-language models or extensive computational resources, remains an open engineering problem.

Second, sexism detection is an inherently subjective task, and annotators frequently disagree on whether content is sexist, and, if so, in what way [3], [4]. Systems trained on majority-vote labels discard this disagreement, producing overconfident predictions on genuinely ambiguous content. While training directly on soft label distributions preserves this information, it is not yet clear how to combine this approach with heterogeneous modalities and annotator demographic information within a single, jointly trained architecture.

Third, not all disagreement between annotators reflects genuine ambiguity in the content; some of it results from annotator inconsistency, spamming, or error [7], [8]. Distinguishing between these two sources of disagreement requires estimating the reliability of individual annotators while preserving systematic disagreement as signal, as proposed by more recent Bayesian approaches [9]. However, models that learn annotator identity directly cannot generalize to annotators unseen during training [10], which limits their applicability once new annotators contribute to a dataset.

This dissertation is motivated by the need to address these three problems jointly, within a single system capable of detecting sexism across memes and short-form video while preserving the subjectivity inherent to the task and distinguishing genuine disagreement from annotation noise.

1.3 Objectives

The main objective of this work is to develop and validate a multimodal architecture for sexism detection in memes and short-form videos, trained under the Learning with Disagreement paradigm and capable of distinguishing genuine annotator disagreement from annotation noise. To achieve this main objective, the following specific objectives were established:

- **Design a multimodal architecture for meme classification** that jointly encodes textual and visual content, and injects annotator demographic information to capture how content is perceived across different population groups;

- **Extend this architecture** to short-form video by integrating visual, textual, auditory, and physiological signals within a shared fusion strategy, allowing the system to exploit modality-specific evidence while remaining robust to missing modalities;

- **Train both systems directly on soft label distributions** rather than majority-vote labels, so that model predictions reflect the level of agreement or disagreement present in the original annotations across all three prediction subtasks (binary detection, source intention, and fine-grained category);

- **Develop an annotator reliability layer** that estimates the consistency of individual annotators during training and generalizes this estimate to annotators unseen at inference time, through demographic-conditioned mapping rather than fixed annotator identities;

- **Validate the proposed system** through experiments on meme and video datasets annotated under this scheme, evaluating its performance against unimodal baselines and majority-vote-trained systems, and assessing the individual contribution of each architectural component through ablation studies.

This approach focus on improving the robustness and calibration of sexism detection systems by combining modalities and preserving the subjectivity of the annotation process, instead of collapsing it into a single decision. The proposed system addresses key limitations of traditional single-modality and majority-vote-based approaches, including their inability to capture cross-modal meaning, their tendency to produce overconfident predictions on ambiguous content, and their limited capacity to generalize annotator reliability estimates beyond the specific individuals seen during training.

1.4 Document Structure

This dissertation is organized into eight chapters and one appendix that progressively develop the proposed multimodal system from theoretical foundations to practical implementation and validation.

Chapter 1: Introduction establishes the research context, motivations, and objectives. It examines the limitations of current sexism detection approaches when faced with multimodal content and subjective annotation, and introduces the Learning with Disagreement paradigm as a promising foundation for the proposed system.

Chapter 2: State of the Art provides a comprehensive review of relevant literature organized into four main areas: sexism and hate speech detection and its evolution towards multimodal approaches; multimodal learning techniques applied to memes and short-form video; the Learning with Disagreement paradigm and soft-label training; and annotator modeling and reliability estimation, identifying gaps that this dissertation addresses.

Chapter 3: Task and Dataset Description describes the problem formulation across its three prediction subtasks, the meme and video datasets used in this work, the annotator demographic metadata, the physiological data sub-corpus, and the evaluation metrics adopted throughout the dissertation.

Chapter 4: Meme System Architecture presents the first core contribution of this work: a detailed specification of the multimodal architecture developed for sexism detection in memes. This chapter describes the text and visual encoders, the fusion strategy adopted after empirically discarding cross-modal attention, the annotator demographic injection mechanism, and the training objectives used to jointly optimize the three prediction subtasks.

Chapter 5: Video System Architecture extends this architecture to short-form video, detailing the video, audio, and text encoders, the integration of eye-tracking and heart-rate physiological signals, the multimodal fusion strategy with modality dropout, and the corresponding training configuration.

Chapter 6: Annotator Reliability Modeling presents the second core contribution of this work: an annotator reliability layer that distinguishes genuine disagreement from annotation noise. This chapter motivates the problem, describes the reliability estimation process used during training, and details the demographic- conditioned mapping developed to generalize this estimate to annotators unseen at inference time.

Chapter 7: Experimental Setup and Results documents the implementation of both proposed architectures, the experimental methodology used for validation, the ablation study conducted to isolate the contribution of individual architectural components, the results of the annotator reliability study, and a comprehensive analysis of the system's performance across both media types.

Chapter 8: Conclusions synthesizes the main contributions of this research, discusses the limitations of the current work, and proposes directions for future research to extend the proposed multimodal, disagreement-aware detection system.

Throughout these chapters, the dissertation progressively builds from the theoretical foundations and state-of-the-art review toward the practical implementation and validation of a novel multimodal system that addresses the challenges of sexism detection across memes and short-form video.

Chapter 2

State of the Art

This chapter presents the state of the art relevant to the development of a multimodal system for sexism detection under the Learning with Disagreement paradigm. The analysis is organized into five complementary sections: Section 2.1 examines the evolution of sexism and hate speech detection, from early text-only methods to current multimodal approaches; Section 2.2 analyzes multimodal learning architectures applied to memes and short-form video, together with their limitations; Section 2.3 introduces the Learning with Disagreement paradigm and soft-label training as an alternative to majority-vote labeling; Section 2.4 reviews annotator modeling and reliability estimation techniques, from early competence-estimation models to recent demographic-aware approaches; and Section 2.5 synthesizes these four perspectives, identifying the gaps that this dissertation aims to address.

2.1 Sexism and Hate Speech Detection

Sexism and hate speech detection has evolved over the past decade, following the broader progression of natural language processing and computer vision techniques. Understanding this evolution is essential for situating system proposed in this dissertation.

Early approaches to sexism and hate speech detection relied entirely on textual content. [1] framed hate speech detection as a supervised classification problem over surface-level linguistic features, such as character n-grams and word embeddings, applied to short social media posts. These text-only methods proved effective for explicit forms of hate speech, characterized by unambiguous slurs or overtly aggressive language, but struggled to capture more implicit or context-dependent expressions, where meaning depends on cultural references, sarcasm, or visual elements absent from the text itself.

The introduction of multimodal benchmarks revealed the limitations of unimodal approaches for social media content in which discriminatory meaning arises specifically from the interaction between text and image. [2] demonstrated, through the Hateful Memes Challenge, that meme classification systems relying solely on text or solely on image features consistently underperform relative to systems that jointly reason over both modalities, since a meme's message frequently depends on the juxtaposition of an image with an unrelated or contradictory caption. This finding established multimodal reasoning as a necessary, rather than optional, component of hateful and sexist content detection in visual social media formats.

More recently, the object of detection has extended beyond static images to short-form video, a format that introduces additional temporal and auditory dimensions. [11] proposed one of the first multimodal frameworks specifically targeting sexism detection in TikTok videos, combining textual transcripts, audio features, and visual frames, and showed that this combination substantially outperforms text-only baselines on video content, where tone of voice, background music, and visual context frequently carry sexist meaning that is entirely absent from the spoken or written text. Complementing this direction, [12] introduced a Spanish-language video dataset annotated with modality-specific labels, explicitly measuring the contribution of each modality to human sexism judgments. Their results show that visual cues play a decisive role in the recognition of implicit or indirect sexism, cases in which annotators relying on the text transcript alone frequently disagree with annotators who have access to the full video, underscoring both the necessity of multimodal analysis and the inherently subjective nature of the task.

Taken together, this body of work traces a clear trajectory from unimodal, explicit hate speech detection towards multimodal, context-dependent sexism characterization across increasingly rich media formats. This trajectory motivates a closer examination of the multimodal learning architectures that make such joint reasoning possible, which is the focus of the next section.

2.2 Multimodal Learning for Memes and Video

Building on the need for cross-modal reasoning established in the previous section, this section examines the architectures developed to jointly process textual, visual, and, for video content, auditory information, along with the specific limitations that motivate the architectural choices adopted in this dissertation.

Contrastive vision-language pre-training has become a foundational technique for representing images and text within a shared embedding space. [13] introduced CLIP, a model trained on large-scale image-text pairs using a contrastive objective, producing visual and textual encoders whose representations are directly comparable through cosine similarity. While CLIP was designed as a general-purpose vision- language model rather than one tailored to hateful or sexist content, its representations have become a common starting point for multimodal meme classification, given the strong alignment they provide between image and text semantics without requiring paired supervision specific to the target task.

Building on this foundation, [14] proposed Hate-CLIPper, which explicitly models cross-modal interactions between CLIP-derived image and text features through a feature interaction matrix, rather than treating them as independent inputs to be concatenated. This architecture specializes the general-purpose alignment learned by CLIP into a task-specific representation for hateful meme detection, and established a strong baseline for subsequent multimodal meme classification systems. More recent submissions to sexism detection shared tasks have explored an alternative direction, leveraging large pretrained multimodal models directly through fine-tuning or prompting rather than designing specialized fusion mechanisms [15]. Both directions, however, share a common limitation for the sexism detection setting: neither explicitly models the annotation disagreement that characterizes sexism as a subjective task, treating each meme as having a single correct label to be predicted.

Extending multimodal reasoning to short-form video introduces additional architectural challenges, since video content combines temporal, visual, textual, and auditory signals that must be aligned and fused. [5] proposed VideoMAE, a self-supervised video encoder trained via masked autoencoding, which learns effective spatio-temporal representations from a comparatively small number of labeled videos, making it particularly suitable for tasks such as sexism detection where large annotated video corpora are not available. For the auditory component, [16] introduced Whisper, a speech recognition and audio representation model trained on large-scale weakly supervised data, robust to the noisy, non-studio audio conditions typical of user- generated video content. Combining video, audio, and text encoders into a single system requires a fusion strategy capable of handling the resulting high-dimensional, heterogeneous representation space, as well as robustness to cases where one modality is degraded or entirely missing, a common occurrence in real-world video content where audio may be silent, absent, or uninformative.

A recurring design decision across this literature concerns how modality-specific representations should be combined. Approaches based on cross-modal attention allow each modality to dynamically weigh evidence from the others, offering, in principle, a more expressive fusion mechanism. In practice, however, such mechanisms require substantial training data to learn meaningful attention patterns, and can collapse to near-uniform weighting when trained on comparatively small, specialized datasets such as those used for sexism detection. Simpler fusion strategies, such as concatenating pooled representations from each modality before a shared classification head, have been shown in several settings to match or exceed the performance of more elaborate attention-based mechanisms under these data constraints, while being considerably lighter to train and less prone to overfitting.

This trade-off between architectural expressiveness and data efficiency directly informs the fusion strategies adopted for both the meme and video systems developed in this dissertation, detailed in Chapters 4 and 5, respectively. The next section turns to a distinct but complementary challenge: how these multimodal systems should be trained when the labels themselves reflect genuine disagreement among annotators.

2.3 Learning with Disagreement and Soft-Label Training

The previous sections established that sexism detection requires multimodal reasoning and that annotators frequently disagree on whether content is sexist. This section explains how this disagreement can be explicitly incorporated into model training, rather than discarded through majority vote.

The conventional approach to supervised classification collapses multiple annotator judgments into a single label, typically via majority vote, treating any remaining disagreement as noise to be resolved rather than as signal to be modeled. [10] show that this practice discards systematic information: for genuinely subjective tasks, disagreement often correlates with identifiable factors such as the content's ambiguity or the annotators' own backgrounds, rather than reflecting random labeling error. Collapsing this variation into a single label therefore produces models that are confidently wrong on precisely the instances where human judges themselves were uncertain or divided.

The Learning with Disagreement (LeWiDi) paradigm addresses this limitation by reframing the prediction target itself. Instead of predicting a single aggregated label, a model trained under this paradigm predicts the full distribution of annotator judgments for each instance, commonly referred to as a soft label. [3] survey this shift across NLP and computer vision tasks, arguing that treating disagreement as signal, rather than noise, is particularly important for inherently subjective tasks such as sexism, hate speech, and irony detection, where no single ground truth may exist independently of the annotator's perspective.

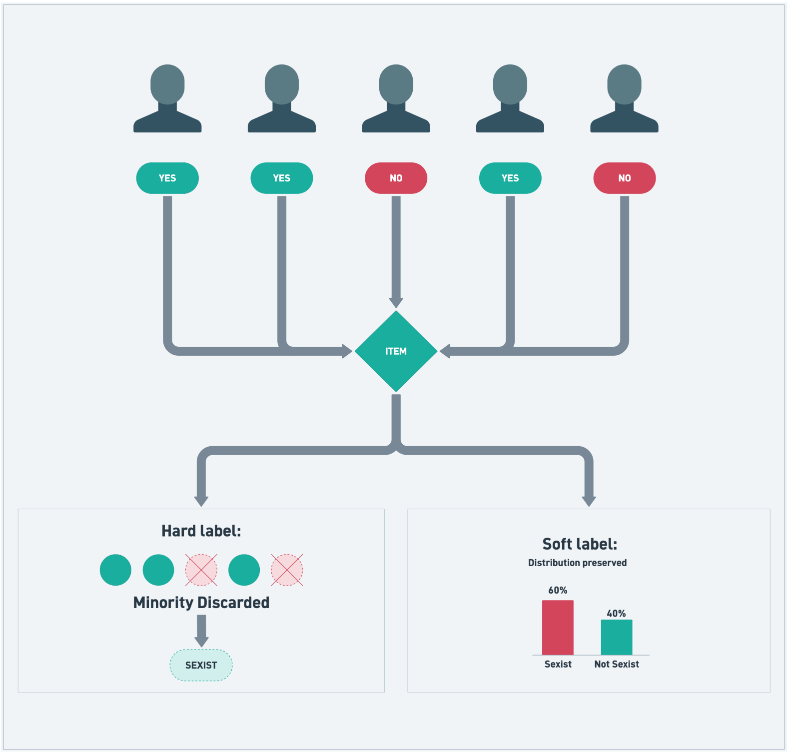

Figure 2.1: Conceptual comparison between hard-label and soft-label training. Five annotators cast individual judgments on the same content; the hard-label approach collapses this into a single majority-vote label and discards the minority votes, while the soft-label approach preserves the full distribution of annotator judgments.

[4] further formalizes this position, characterizing human label variation as a property of the data-generating process itself, rather than as a symptom of poor annotation quality, and arguing that evaluation protocols should be adapted according to that.

The LeWiDi shared task series has served as the primary place for developing and benchmarking soft-label training methods. [17] organized the second edition of this task around subjective NLP classification problems, training and evaluating systems directly against soft-label distributions rather than hard labels, and reporting that directly optimizing for the target distribution, typically via a distributional loss such as cross-entropy or Manhattan distance between predicted and observed label distributions, consistently outperforms training on aggregated labels and subsequently calibrating the output. The third edition of this task, [18], further distinguishes between two complementary evaluation paradigms: predicting the population-level distribution of judgments for an instance, and predicting the specific interpretation of an individual annotator, a distinction that is directly relevant to the annotator reliability modeling developed in this dissertation.

Physiological signals collected alongside subjective annotations represent a further extension of this paradigm, offering a potential behavioral correlate of the perceptual and affective processes underlying disagreement itself, an avenue that remains largely unexplored in the sexism detection literature [6].

Training directly on soft labels, rather than majority-vote labels, is the approach adopted throughout the meme and video systems developed in this dissertation, detailed in Chapters 4 and 5. However, soft-label training alone does not distinguish between two fundamentally different sources of the disagreement it preserves: genuine ambiguity in the content, and inconsistency or unreliability on the part of individual annotators. This distinction is the focus of the next section.

2.4 Annotator Modeling and Reliability Estimation

The previous section established that training directly on soft labels preserves annotator disagreement as informative signal. However, not all of this disagreement reflects genuine ambiguity in the content: some of it results from annotator inconsistency, carelessness, or error. This section reviews the evolution of techniques for estimating annotator reliability, from early statistical models to recent demographic-aware approaches, motivating the annotator reliability studies developed in this dissertation.

The foundational approach to modeling annotator reliability was proposed by [8], who introduced an Expectation-Maximization algorithm for jointly estimating the true label of each item and the error rate of each annotator from multiple noisy annotations, without requiring access to ground truth during training. This formulation established the basic premise that underlies most subsequent work in this area: annotators are not equally reliable, and this reliability can be estimated directly from patterns of agreement and disagreement within the annotation data itself.

[7] extended this idea with MACE, a model that explicitly distinguishes between annotators who are honest but occasionally noisy and annotators who behave as spammers, assigning a competence score to each annotator based on how consistently their labels align with the inferred consensus. [19] later provide a systematic comparison of Bayesian models of annotation, including Dawid-Skene and MACE-style formulations, showing that the choice of model has a measurable impact on the quality of the inferred ground truth, particularly under sparse or unbalanced annotation patterns, a condition common in large-scale crowdsourced datasets.

These early reliability models, while effective at identifying unreliable annotators, were not designed to distinguish reliability from genuine, systematic disagreement rooted in annotator perspective. [10] argue that collapsing annotator identity into a single competence score can obscure meaningful patterns of disagreement that correlate with annotator background, and propose multi-task architectures that predict each annotator's individual judgment rather than a single aggregated reliability estimate. This shift towards perspective-aware modeling motivated subsequent work that incorporates annotator identity or demographic information directly into the learned representation. [20] introduce jury learning, which models every annotator individually and composes configurable juries with a specified demographic composition, allowing practitioners to control which perspectives are reflected in the final prediction. [21] propose DEM-MoE, which routes predictions through expert subnetworks conditioned on annotator demographics, explicitly capturing group-level and intersectional variation in a more structured form than individual annotator embeddings.

A common limitation across these embedding-based approaches is that they learn representations tied to the specific annotators seen during training, and therefore cannot generalize to annotators who did not contribute to the training set. [22] address this parametric efficiency and generalization bottleneck by replacing fixed annotator embeddings with hypernetwork-generated adapters, allowing the model to adapt to new annotators without retraining from scratch. This limitation is particularly relevant for annotator reliability modeling: a system that only recognizes annotators present during training cannot estimate the reliability of new annotators contributing to a dataset after deployment, restricting its practical applicability. [9] propose NUTMEG, a more recent approach that separates genuine signal from noise in annotator disagreement while explicitly preserving systematic, perspective-driven disagreement rather than treating all variation as a symptom of unreliability, addressing a key shortcoming of earlier competence-based models. [23] take a complementary, active-learning perspective, proposing annotator-centric selection strategies that account for individual annotator characteristics when deciding which items and which annotators to prioritize for further labeling in subjective NLP tasks.

Taken together, this progression reveals a trajectory from purely statistical competence estimation, through individual annotator embeddings, towards demographic-conditioned approaches capable of generalizing beyond the specific annotators observed during training. This last property, generalization to unseen annotators via demographic conditioning, rather than fixed annotator identity, is the central design principle adopted by the annotator reliability layer developed in this dissertation, detailed in Chapter 6.

2.5 Towards Demographic-Aware, Disagreement-Preserving Systems

This section synthesizes the four perspectives reviewed in this chapter, allowing us to identify the specific gaps that this dissertation addresses. The analysis is structured around two complementary axes: first, the extent to which a system integrates multiple modalities rather than relying on text alone; second, the extent to which a system preserves annotator disagreement as prediction target rather than collapsing it into a majority-vote label.

Multimodal learning has substantially advanced the detection of hateful and sexist content in memes and, more recently, in short-form video, as reviewed in Sections 2.1 and

2.2. Architectures built on contrastive vision-language encoders and specialized cross-modal fusion mechanisms consistently outperform unimodal baselines when discriminatory meaning arises from the interaction between modalities [2], [14]. However, these systems are almost universally trained on majority-vote labels, treating sexism as a task with a single correct answer per instance, despite the inherently subjective nature of the judgment being modeled.

Conversely, the Learning with Disagreement paradigm, reviewed in Section 2.3, has demonstrated that training directly on soft-label distributions produces better-calibrated predictions on genuinely ambiguous content [3], [17]. However, this line of work has developed almost entirely within unimodal, text-based NLP tasks, and its extension to multimodal settings, where disagreement may arise jointly from visual, textual, and auditory ambiguity, remains comparatively unexplored.

A third, largely independent line of work, reviewed in Section 2.4, has focused on modeling annotator reliability and demographic background directly, moving from early competence-estimation models towards demographic-conditioned architectures capable of generalizing to annotators unseen during training [21], [22]. This work has, so far, been developed independently of both the multimodal fusion techniques discussed in Section 2.2 and the soft-label training objectives discussed in Section 2.3, rather than as an integrated component of a single system.

The intersection of these three lines of work, illustrated in Figure 2.2, remains largely unexplored. Existing multimodal systems for sexism and hate speech detection predominantly occupy the majority-vote quadrant of this landscape, regardless of how many modalities they integrate. Systems trained under the Learning with Disagreement paradigm predominantly remain unimodal. Demographic-aware annotator reliability models, in turn, have not yet been integrated into multimodal, soft-label-trained architectures for sexism detection specifically, nor extended to short-form video content enriched with physiological signals [6].

This dissertation proposes to address this gap by developing a multimodal system

Figure 2.2: Research landscape for sexism detection approaches.

for sexism detection in memes and short-form video that is trained directly on soft-label distributions and incorporates a demographic-conditioned annotator reliability layer capable of generalizing to unseen annotators. The system leverages the cross-modal fusion strategies identified in Section 2.2 as most effective under limited training data, while adopting the soft-label training objective established in Section 2.3, and extends the demographic-aware reliability modeling reviewed in Section 2.4 into a jointly trained, multimodal setting. Chapters 4 and 5 detail the resulting architecture for memes and video, respectively, and Chapter 6 details the annotator reliability layer that completes this integration.

Chapter 3

Task and Dataset Description

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3.3 Video Dataset

3.4 Annotator Demographic Metadata

3.5 Physiological Data Sub-Corpus

3.6 Evaluation Metrics (ICM-Norm, Soft vs. Hard)

3.7 Final Considerations

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4.2 Text and Visual Encoders

4.3 Fusion Strategy

4.4 Annotator Demographic Injection

4.5 Training Objectives and Optimization

4.6 Per-Class Threshold Tuning

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Video System Architecture (Task 3)

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5.2 Video, Audio, and Text Encoders

5.3 Biometric Feature Integration

5.4 Multimodal Fusion and Modality Dropout

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Chapter 6

Annotator Reliability Modeling

6.1 Motivation: Noise vs. Genuine Disagreement

6.2 Reliability Estimation During Training

6.3 Demographic-Conditioned Generalization to Unseen Annotators

6.4 Integration with the Multimodal Pipeline

Chapter 7

Experimental Setup and Results

7.1 Experimental Setup (Environment, Seeds, Ablation Design)

7.2 Meme System Results

7.3 Video System Results

7.4 Ablation Study

7.5 Annotator Reliability Study Results

7.6 Results Discussion

Chapter 8

Conclusions

8.1 Summary of Contributions

8.2 Limitations of the Study

8.3 Directions for Future Work

8.4 Final Considerations

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Appendix A

Appendix

A.1 Implementation Availability